SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics	**COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5 Duration: 60 Minutes

Code of Conduct

I **B.V. THARUN (192525155)** (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

Assessment Tasks
Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

- A hypervisor is the software layer that creates and runs virtual machines (VMs) by dividing the physical resources of a server (CPU, memory, storage, network). A **Type-1 (bare-metal)** hypervisor is installed directly on the physical hardware. A **Type-2 (hosted)** hypervisor is installed as an application on top of a host operating system.

Comparison of Type-1 and Type-2 hypervisors:

Parameter	Type-1 (Bare-metal)	Type-2 (Hosted)
Architecture	Hardware → Hypervisor → VMs	Hardware → Host OS → Hypervisor → VMs
Performance	Direct access to CPU, memory and I/O, so overhead is very low and performance is near-native. Supports hardware-assisted virtualization and large VM density.	Every request passes through the host OS, adding an extra layer, so latency and overhead are higher. Host OS also competes for resources.
Security	Small code base (small attack surface). No general-purpose OS, no browsers or user applications. Strong VM isolation and secure boot support.	Large attack surface because it inherits the host OS vulnerabilities, drivers, malware and user applications. A host OS compromise exposes all VMs.
Manageability	Centralized management of many hosts (clusters, live migration, HA, automated resource scheduling, monitoring and policy control).	Managed per machine through a desktop interface. Limited clustering, no native HA or large-scale automation.
Reliability	High availability and fault tolerance features.	Depends on the host OS stability; host reboot or patching stops all VMs.
Examples	VMware ESXi, Microsoft Hyper-V, Xen, KVM	VMware Workstation, Oracle VirtualBox, Parallels
Typical use	Cloud datacenters and enterprise servers	Testing, development, training and desktops

Analysis:

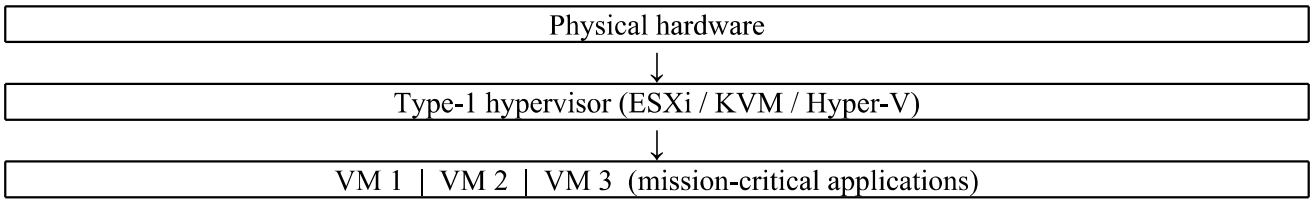
Performance: mission-critical applications need predictable low latency and high throughput, and because Type-1 removes the host OS layer it gives better CPU scheduling, memory management and I/O. **Security:** fewer components mean fewer vulnerabilities, so Type-1 isolates tenants more strongly, while Type-2 depends on the security of the host operating system. **Manageability:** Type-1 supports live migration, high availability, load balancing and automation, which are essential for managing thousands of VMs in a datacenter.

Justification of the better choice:

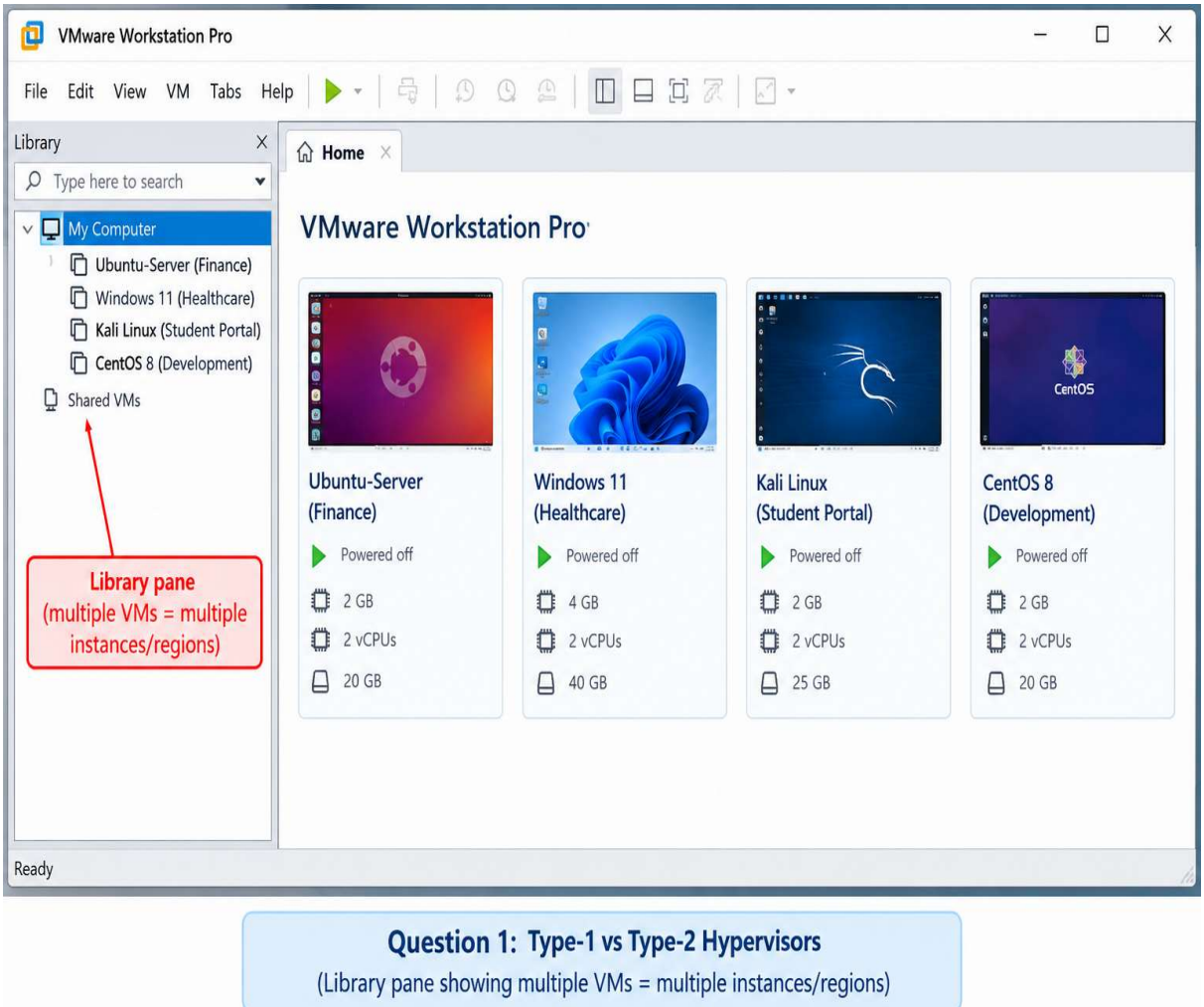
Type-1 hypervisor is the better choice for a cloud datacenter hosting mission-critical applications because it provides higher performance, a smaller attack surface, stronger isolation, and enterprise-level manageability and availability. Type-2 is suitable only for development, testing and learning.

Conclusion: Type-1 hypervisor is the most suitable architecture for a mission-critical cloud datacenter because it combines near-native performance, strong security and enterprise-grade management.

Architecture flow:



VMware Workstation Pro:



Question 1: Type-1 vs Type-2 Hypervisors
(Library pane showing multiple VMs = multiple instances/regions)

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2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

A) The three portals belong to different tenants with different compliance needs (finance: PCI-DSS, healthcare: HIPAA, student: privacy of education records). They share the same physical host, so the virtualization layer must make each workload behave as if it runs on its own separate machine (multi-tenancy with isolation).

How virtualization architecture supports isolation:

VM-level isolation: each portal runs in its own VM with its own guest OS, kernel and applications, so a crash or compromise inside one VM does not directly affect another. **Hypervisor mediation and memory isolation:** the hypervisor controls all access to CPU, memory and devices, so a VM cannot reach the hardware directly, and it uses separate memory address spaces (hardware-assisted paging such as Intel EPT / AMD NPT) so that one VM cannot read another VM's memory. **Network and storage isolation:** separate virtual switches, VLAN / VXLAN segments, virtual firewalls and security groups keep finance, healthcare and student traffic apart, while separate virtual disks, datastores or LUNs with access control and encryption of data at rest are used for each tenant. **Resource isolation and placement:** CPU, memory and I/O reservations, limits and shares prevent the "noisy neighbour" problem, and policy-based placement using anti-affinity rules or dedicated clusters/hosts protects highly regulated workloads such as healthcare and finance.

Workload	Recommended isolation
Finance portal	Dedicated VLAN, encrypted disks, strict firewall rules, audit logging
Healthcare portal	Dedicated host/cluster if possible, encrypted storage and network, access control and logging for patient data
Student portal	Separate VM and network segment with resource limits, lower sensitivity but still isolated

Two major attack surfaces:

Hypervisor and management plane. The hypervisor is a single point of control. A vulnerability can allow a **VM escape** (an attacker breaks out of one VM to the hypervisor and then reaches other tenants) or a hypervisor takeover. The management console, APIs and hypercalls are also targets for unauthorized access.

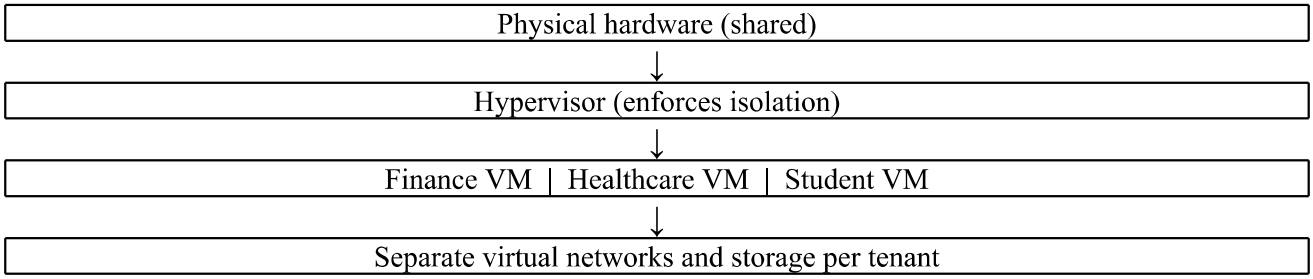
Mitigation: patch hypervisor regularly, use a minimal Type-1 hypervisor, secure boot, role-based access control and MFA for the management interface, and separate the management network.

Shared hardware and virtual network. Tenants share CPU caches, memory and the virtual switch. Attackers can use **side-channel attacks** (cache timing, Spectre/Meltdown type) to steal data across VMs, or use **VLAN hopping, ARP/MAC spoofing and traffic sniffing** on the virtual network. VM images and snapshots that are not encrypted can also leak data.

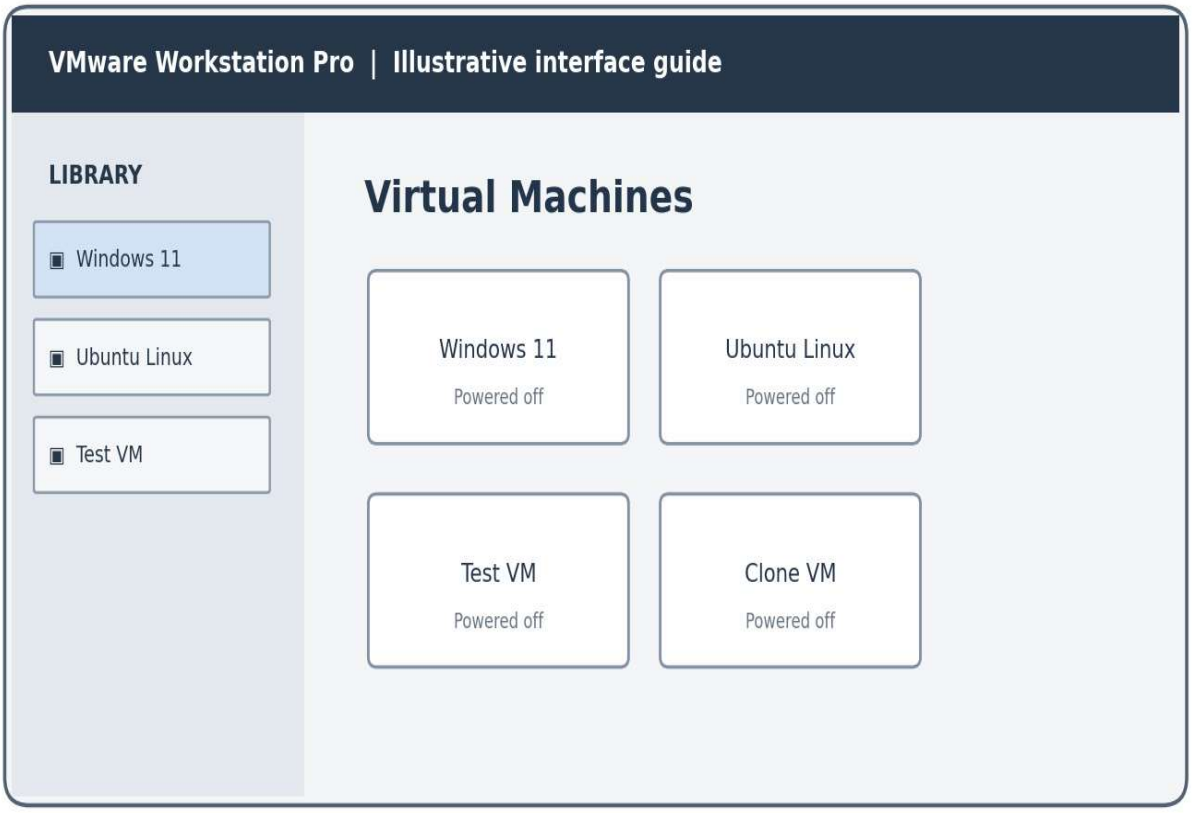
Mitigation: CPU microcode and hypervisor patches, dedicated hosts for sensitive tenants, micro-segmentation, virtual firewalls, encrypted VM disks and snapshots, and network traffic monitoring.

Conclusion: the hypervisor, together with virtual networks, separate storage and resource controls, isolates the three portals on shared hardware. Securing the hypervisor and the shared resources is the key to keeping that isolation reliable.

Isolation model:



VMware Workstation Pro:



3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

A) The snapshot shows that the host is overloaded on all major resources at the same time. The values are compared below with commonly used guideline levels (healthy CPU is generally below about 70-80%, memory below about 80-85%, storage latency below about 10-20 ms, and VM boot of roughly 30-60 s).

Metric	Observed	Healthy guideline	Interpretation
CPU utilization	92%	< 75-80%	Near saturation. VMs wait for CPU time (high CPU ready / contention), so CPU is over-committed.
Memory utilization	88%	< 80-85%	High pressure. Hypervisor may use ballooning, compression or swapping, which slows VMs and adds disk activity.
Storage I/O latency	35 ms	< 10-20 ms	Storage is slow, caused by too many VMs sharing the datastore, swapping or slow disks.
Average VM boot time	170 s	30-60 s	Very slow. Result of CPU and memory contention and high disk latency (boot storm).

Likely bottlenecks (cause-effect analysis):

CPU bottleneck: too many vCPUs are allocated on the host (over-commitment) and some VMs are oversized, so VMs queue for physical CPU. **Memory bottleneck:** memory is almost full, so the hypervisor reclaims it by ballooning or swapping to disk. **Storage I/O bottleneck:** this swap activity, together with normal VM traffic, overloads the shared datastore, so latency rises to 35 ms. **Boot time effect:** booting needs CPU, memory and many disk reads, so with all three constrained the boot time reaches 170 s, and simultaneous VM startups (boot storm) make it worse.

Root cause chain: high memory usage → swapping → more disk I/O → high latency → slow VM boot, together with CPU contention that slows every operation.

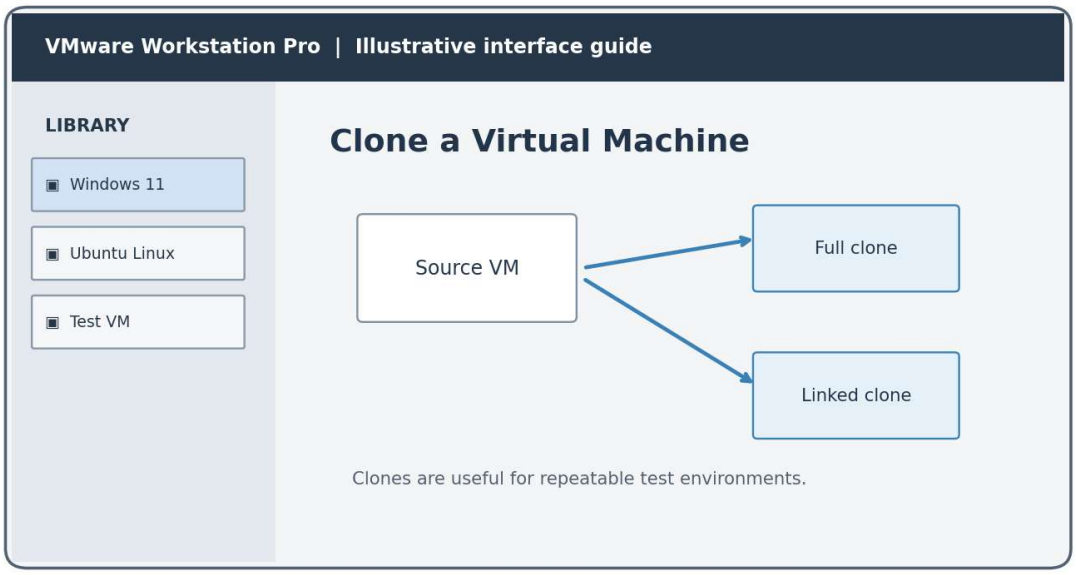
Corrective actions:

Area	Recommended action
CPU	Right-size VMs (reduce unused vCPUs); use live migration (vMotion) / DRS to move VMs to less loaded hosts; add hosts to the cluster; set CPU limits/shares for non-critical VMs.
Memory	Reduce VM memory over-allocation; enable memory page sharing/compression; reserve memory for critical VMs; add RAM or move VMs to other hosts.

Storage	Move disks to SSD/NVMe or a faster tier; spread VMs across multiple datastores; use storage I/O control; enable caching; avoid swapping to slow disks.
Boot time	Stagger VM power-on; use templates or linked clones; keep boot images on fast storage; remove unnecessary startup services.
Monitoring	Use monitoring and alerts (CPU ready, ballooning, datastore latency); set thresholds; plan capacity and use auto-scaling.

Conclusion: the host is overloaded (resource contention). Load balancing, right-sizing, adding capacity and faster storage will bring the utilization and boot time back to normal.

VMware Workstation Pro:



4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

A) A **Software Defined Datacenter (SDDC)** is a datacenter in which all infrastructure (compute, storage, networking and security) is virtualized and delivered as a service, and the entire environment is controlled by software through policies, automation and APIs. In a **traditional datacenter**, hardware is dedicated, configured manually and managed separately by silos (server, storage and network teams).

Traditional datacenter vs SDDC:

Parameter	Traditional datacenter	Software Defined Datacenter
Provisioning	Days or weeks (order, install, cable, configure)	Minutes (self-service, templates, automation)
Control	Hardware-specific, manual	Software-based, policy-driven and API-driven
Scalability	Add physical devices; slow	Scale up or out by software; elastic
Resource utilization	Low (dedicated servers)	High (pooled and shared resources)
Cost	High CAPEX and operations	Commodity hardware, lower OPEX
Management	Separate tools per silo	Single unified management and orchestration
Change and recovery	Slow, error-prone	Fast, using snapshots, templates and automated failover

Virtualization considerations:

Compute virtualization: the hypervisor abstracts the physical servers into VMs (or containers) that are created, cloned, moved (live migration) and scaled in minutes. Resource pools allow dynamic CPU and memory allocation, which improves agility and utilization.

Storage virtualization: physical disks are pooled and presented as virtual volumes (software-defined storage such as vSAN). Policies define capacity, performance and protection; thin provisioning, snapshots and tiering are done in software, so storage is allocated on demand without manual LUN configuration.

Network virtualization: networks are created in software (virtual switches, VLAN/VXLAN overlays, virtual routers, firewalls and load balancers; SDN controller). New networks, subnets and security rules are created without re-cabling, and micro-segmentation improves security.

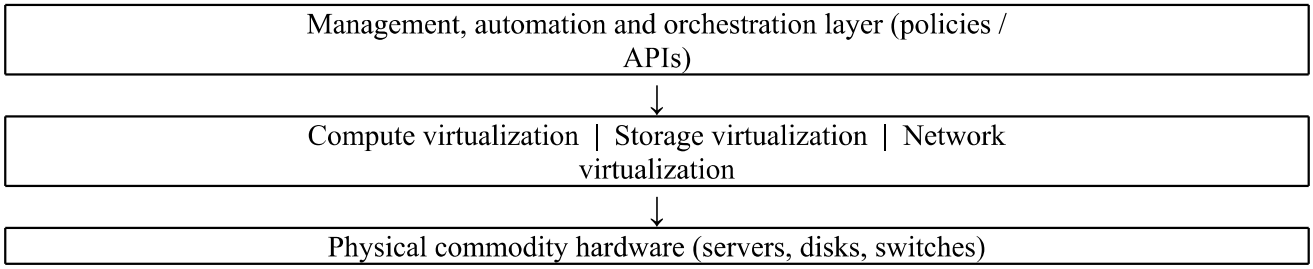
How agility improves:

Agility improves in several ways. Automation and orchestration allow applications and environments to be deployed faster, and the infrastructure can be scaled up or down quickly as demand changes (elasticity). Templates and policies keep configuration consistent and reduce human error, while

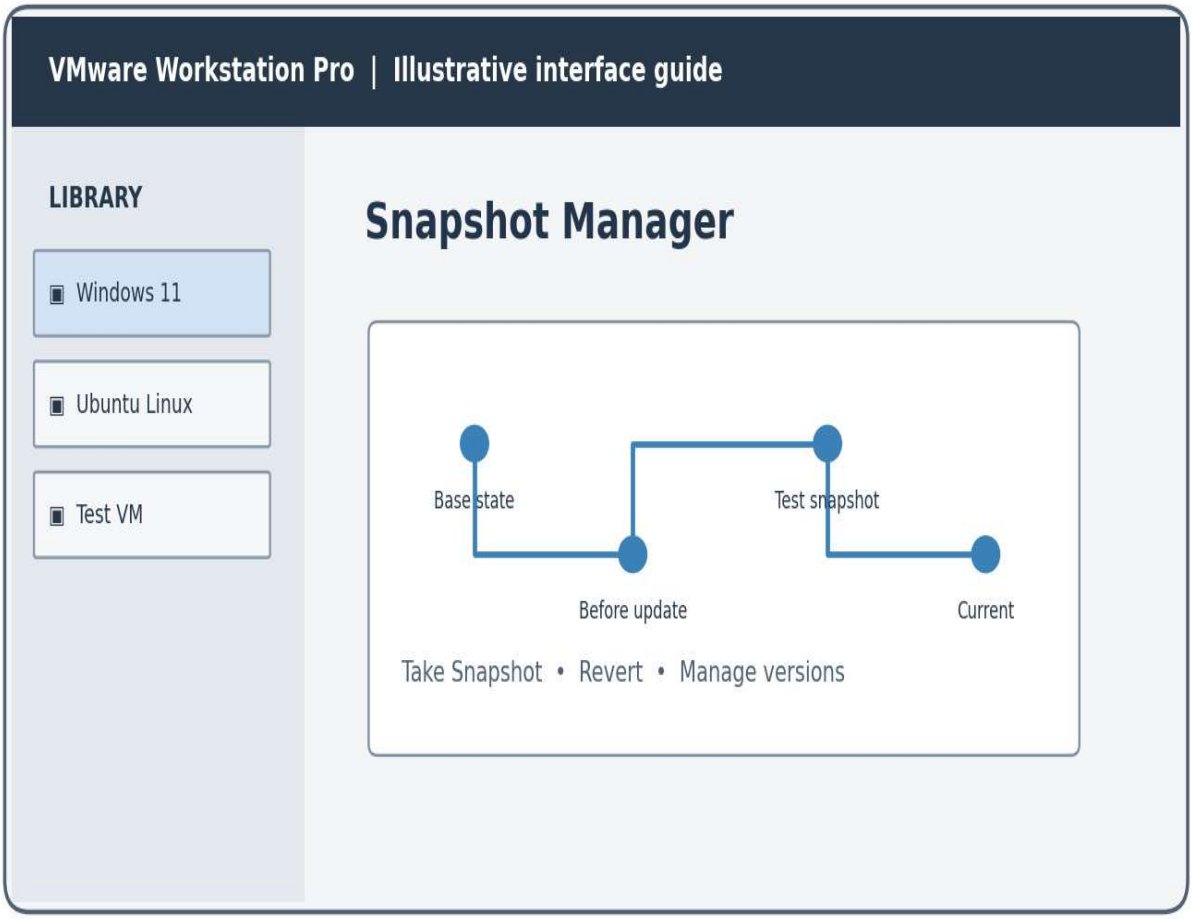
snapshots, replication and live migration make backup, migration and disaster recovery easier. Because the hardware is pooled and vendor-independent, utilization is better and the cost is lower.

Conclusion: by virtualizing compute, storage and network and controlling them through software, an SDDC provisions, scales and recovers infrastructure far faster than a traditional datacenter, giving much higher agility.

SDDC architecture:



VMware Workstation Pro:



5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

A) In a multi-cloud federation, several cloud providers cooperate and share services. Each provider usually has its own identity system (user IDs, roles, attribute names, authentication method and token format). When these are not aligned, a user authenticated in one cloud is not recognized or correctly authorized in another, so the federation fails.

Analysis of the identity mismatch problem:

Cause of mismatch	Effect
Different identity providers and separate user directories	The same user has different accounts or IDs in each cloud; no single identity.
Different protocols and token formats (SAML, OAuth2/OIDC, proprietary)	Tokens issued by one cloud are not understood by another; SSO fails.
Different attribute and role naming (e.g. "admin" vs "Owner")	Wrong or denied access (authorization errors) or excessive privileges.
No common trust relationship or metadata exchange	Providers do not trust each other's assertions or certificates.
Inconsistent policies, time sync and token lifetime	Token rejection, expired sessions, security gaps.
Privacy conflicts (sharing too many attributes, different regional laws)	Data leakage and compliance risk (GDPR, etc.).

Proposed secure identity and privacy design (federation concepts):

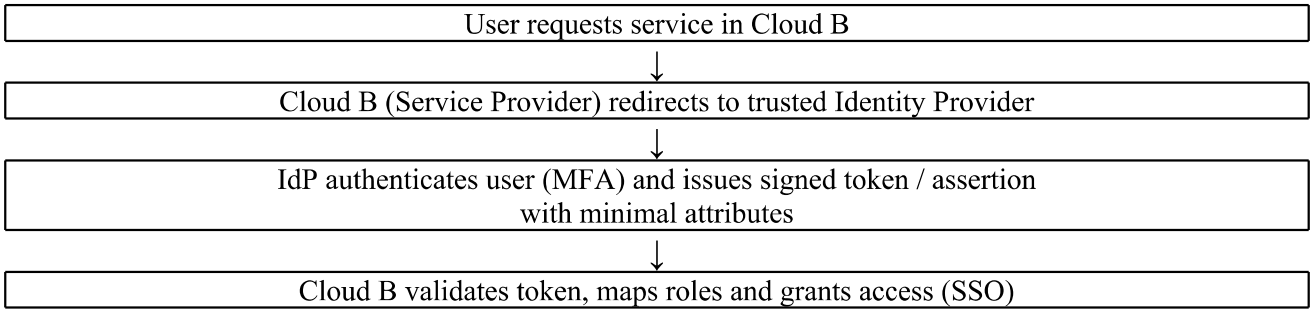
Trusted identity provider / identity broker: a central or trusted Identity Provider (IdP) is used, each cloud acts as a Service Provider (SP), and a federation trust is created by exchanging metadata and certificates. **Standard protocols:** SAML 2.0 or OpenID Connect / OAuth 2.0 is used so that all providers understand the same assertions and tokens, giving Single Sign-On (SSO) across clouds. **Common identity schema and attribute mapping:** a unified user ID is defined and the roles and attributes of each provider are mapped to common roles (RBAC/ABAC), with SCIM used for automatic user provisioning and de-provisioning.

Strong authentication and least privilege: multi-factor authentication (MFA), short-lived signed tokens, token validation and time synchronization (NTP) are used, and each provider enforces local policies on the mapped roles with only the minimum required permissions. **Privacy protection:** only the minimum attributes needed are released (data minimization), pseudonymous or opaque identifiers

per provider prevent user tracking, user consent is obtained, and data residency rules and regulations such as GDPR are followed.

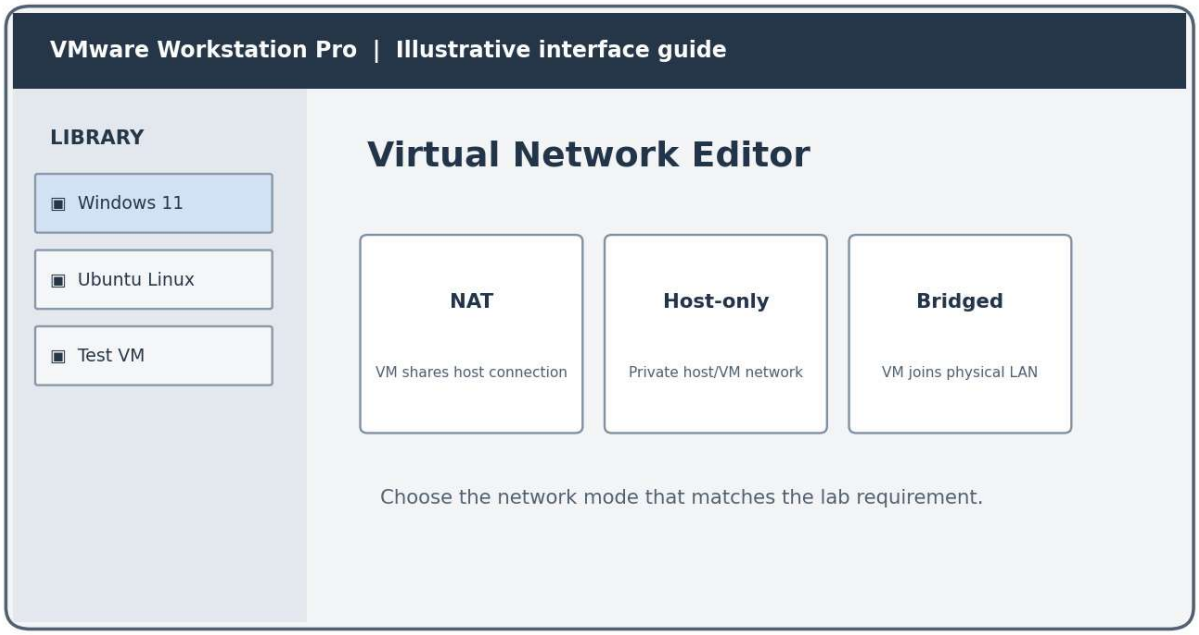
Secure communication and storage: TLS is used for all exchanges, tokens and sensitive attributes are encrypted, and private keys are protected with an HSM or key vault. **Audit, logging and revocation:** authentication events are logged centrally, and tokens or accounts are revoked quickly when a user leaves or a compromise is detected.

Federated identity flow:



Conclusion: the failure is caused by incompatible identities, protocols and attribute models. A federated identity design with a common trust framework, standard protocols, attribute and role mapping, strong authentication and privacy-preserving attribute release solves the mismatch securely.

VMware Workstation Pro:



Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded: